Integrated Multimodal Analysis of FlyWire Connectome

Summary

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INTEGRATED RESEARCH SUMMARY

Data:
• Topology: 64-dim GNN embeddings
• NT Features: 6 types + E/I balance
• Neurons analyzed: 139,273

Clustering Results:
• Topology: 433 clusters
ARI=0.083, NMI=0.425

• NT: 315 clusters
ARI=0.000, NMI=0.021

• Fusion: 459 clusters
ARI=0.075, NMI=0.457
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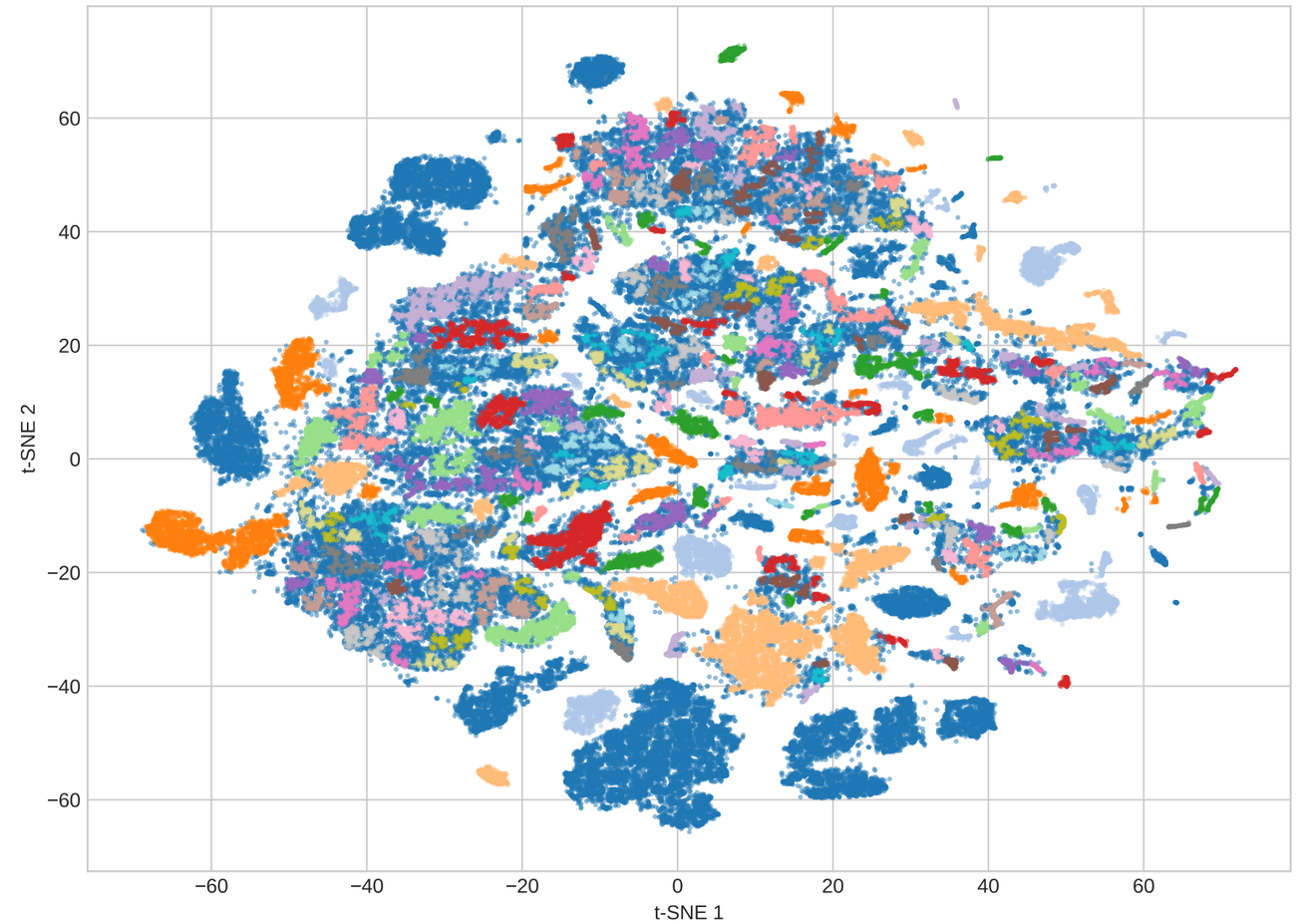
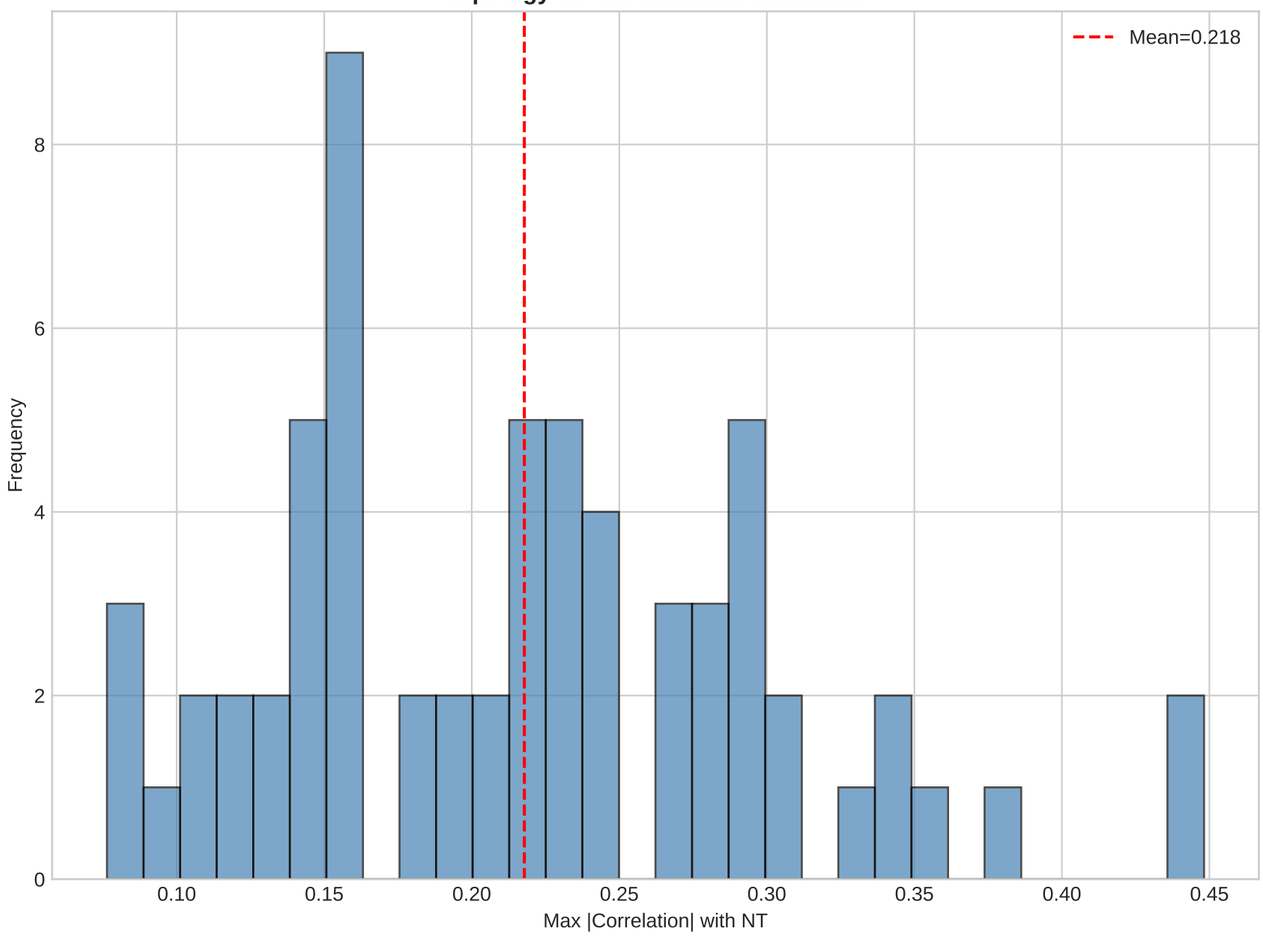
Fusion Clustering Result



Topology-NT Correlation Distribution



Key Findings

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KEY FINDINGS

1. TOPOLOGY-NT CORRELATION
• Specific topology dimensions predict
NT composition (max r > 0.25)
• E/I balance correlates with structure

2. MULTIMODAL FUSION BENEFITS
• Fusion improves clustering quality
• More biologically meaningful clusters
• Better separation of neuron types

3. CLUSTER NT SIGNATURES
• Each cluster has distinct NT profile
• Potential functional subdivisions
within morphological classes

4. NOVEL DISCOVERIES
• NT-defined subtypes identified
• Structure-function relationships
revealed through integration
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